

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct

I M.Mahammad Rafi (192421364) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Mahammad Rafi

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead

transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Assessment Tool 1 – Short Answer Test Solutions

1. Solution

Given:

- File size = 150 MB = 150,000,000 bytes
- Segment size = 1500 bytes
- Data Link Layer overhead = 40 bytes/frame
- Bandwidth = 50 Mbps

Calculation:

Number of segments = $150,000,000 \div 1500 = \mathbf{100,000}$

Number of frames = **100,000**

Total overhead = $100,000 \times 40 = \mathbf{4,000,000 \text{ bytes (4 MB)}}$

Total data transmitted = 150 MB + 4 MB = **154 MB**

Transmission time = $(154 \times 8) \div 50 = \mathbf{24.64 \text{ seconds}}$

2. Solution

Given:

- Sliding window size = 6 frames
- Total frames = 48
- Frame size = 1024 bytes

Calculation:

Number of windows = $48 \div 6 = 8$

One frame is lost in each window.

Total retransmissions = **8 frames**

Flow control regulates the sender's transmission rate according to the receiver's capacity. This prevents data loss, avoids network congestion, and improves communication efficiency by ensuring smooth and reliable data transfer.

3. Solution

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes per fragment

Calculation:

Number of fragments = $12,000 \div 1500 = 8$

Header overhead = $8 \times 20 = 160$ bytes

4. Solution

Given:

- Sliding window size = 6
- Total frames = 48

Calculation:

Number of windows = **8**

Retransmissions = **8 frames**

5. Solution

Given:

- Video size = 600 MB
- Checkpoint after every 60 MB
- Failure after 390 MB

Calculation:

Number of checkpoints before failure = $390 \div 60 = \mathbf{6}$

Last checkpoint = **360 MB**

Data to retransmit = $390 - 360 = \mathbf{30\ MB}$

6. Solution

Given:

- Original file = 900 MB

- Compression = 40%
- Encryption overhead = 5%

Calculation:

Compressed size = $900 \times 60\% = \mathbf{540 \text{ MB}}$

Encrypted size = $540 \times 105\% = \mathbf{567 \text{ MB}}$

Percentage reduction

$$= ((900 - 567) \div 900) \times 100$$

$$= \mathbf{37\%}$$

7. Solution

Given:

- Data size = 3 GB = 3000 MB
- Request size = 3 MB
- Throughput = 120 Mbps

Calculation:

Number of requests = $3000 \div 3 = \mathbf{1000}$

Transmission time

$$= (3000 \times 8) \div 120$$

$$= \mathbf{200 \text{ seconds}}$$

8. Solution

Given:

Application delay = 4 ms

Transport delay = 3 ms

Network delay = 5 ms

Data Link delay = 2 ms

Physical delay = 1 ms

Propagation delay = 18 ms

Transmission delay = 12 ms

Calculation:

Processing delay

$$= 4 + 3 + 5 + 2 + 1$$

$$= \mathbf{15\ ms}$$

Total end-to-end delay

$$= 15 + 18 + 12$$

$$= \mathbf{45\ ms}$$

9. Solution

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead = 80 bytes/frame

Calculation:

Useful data per frame

$$= 1600 - 80$$

$$= \mathbf{1520 \text{ bytes}}$$

Number of frames

$$= 80,000,000 \div 1520$$

$$\approx \mathbf{52,632 \text{ frames}}$$

Total overhead

$$= 52,632 \times 80$$

$$= \mathbf{4,210,560 \text{ bytes}}$$

Transmission efficiency

$$= (1520 \div 1600) \times 100$$

$$= \mathbf{95\%}$$

10. Solution

Given:

- Original file = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Overhead = 60 bytes/frame

Calculation:

Compressed size

$$= 240 \times 70\%$$

$$= \mathbf{168 \text{ MB}}$$

Number of segments

$$= 168,000,000 \div 1400$$

$$= \mathbf{120,000 \text{ segments}}$$

Protocol overhead

$$= 120,000 \times 60$$

$$= \mathbf{7,200,000 \text{ bytes (7.2 MB)}}$$

Final transmitted data

$$= 168 + 7.2$$

$$= \mathbf{175.2 \text{ MB}}$$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand